

The role of air plethysmography in monitoring results of venous surgery

David L. Gillespie, MD, Paul R. Cordts, MD, Choli Hartono, BA, Jonathan Woodson, MD, Eliot Obi-Tabot, MD, Wayne W. LaMorte, MD, PhD, and James O. Menzoian, MD, *Boston, Mass.*

The development of an objective, noninvasive method to assess the hemodynamic effects of venous surgery has long been awaited. Previous methods used to evaluate the results of surgery for varicose veins and venous stasis ulceration have been limited in their quantitative assessment. Now, by use of air plethysmography (APG), we can accurately quantify the effectiveness of corrective venous surgery. Twenty-five extremities that had evidence of venous insufficiency were examined with use of APG before and after venous surgical procedures. Surgery was directed at specific sites of venous incompetence as defined by physical examination and high-resolution duplex imaging. Twenty-one extremities underwent ligation and stripping of the greater saphenous vein. In these patients, APG showed an improvement in venous reflux as demonstrated by a decrease in the venous filling index from 6.6 ± 0.7 ml/sec to 1.8 ± 0.3 ml/sec ($p = 0.0001$) and venous volume from 177.1 ± 14.5 ml to 139.2 ± 8.9 ml ($p = 0.0008$). In addition, these patients showed a mild improvement in calf muscle pump function as noted by an improvement in ejection fraction from $45.8\% \pm 2.0\%$ to $50.8\% \pm 2.5\%$ ($p = 0.07$). The residual volume fraction decreased from $45.0\% \pm 3.4\%$ to $42.0\% \pm 3.7\%$, a difference that was not statistically significant ($p = 0.4$). Four extremities with grade III chronic venous insufficiency underwent popliteal vein valve transplantation with use of an autogenous axillary vein valve. Venous reflux in these patients was reduced dramatically in the early postoperative period, with the venous filling index dropping from 14.2 ± 2.3 ml/sec before the operation to 5.1 ± 1.4 ml/sec after the operation ($p = 0.08$) in conjunction with a drop in venous volume from 190.9 ± 35.4 ml to 135.2 ± 10.7 ml ($p = 0.15$). APG offers a unique and quantitative technique to evaluate the results of reconstructive venous surgery. It is simple and noninvasive, and we believe it represents an improvement on older methods for evaluating the effect of surgery on the venous hemodynamics of the leg. (*J VASC SURG* 1992;16:674-8.)

As our knowledge of the physiologic and anatomic features responsible for lower extremity venous disease expands, we are beginning to recognize the role of corrective venous surgery in its management. Procedures such as ligation and stripping, or flush ligation, of the greater saphenous vein (GSV) have been used for years without a noninvasive, physiologic, or quantitative method to monitor their effectiveness. Invasive determinations of ambulatory venous pressure have been used to examine the results of venous surgery.¹ Attempts have been made

recently to examine the results of standard venous surgical techniques with use of many different noninvasive modalities.^{2,3} The limitations of these methods have been that they do not provide quantitative information to evaluate venous reflux, venous outflow obstruction, and calf muscle pump function. Now, by use of air plethysmography (APG) we should be able to correlate the physiologic alterations in venous function with the anatomic changes as documented by venous duplex scanning. This study attempts to document these changes and give venous surgeons an accurate method to monitor their results.

MATERIAL AND METHODS

Air Plethysmography

Equipment. The APG-1000 (ACI Medical, Sun Valley, Calif.) is a pneumatic plethysmograph designed for measuring absolute volume changes in the

From the Department of Surgery, Boston University School of Medicine, Boston.

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Reprint requests: James O. Menzoian, MD, Chief, Section of Vascular Surgery, Division of Surgery, Boston University Medical Center, 88 E. Newton St., Boston, MA 02118.

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lower leg. Patients are fitted with a 14-inch polyurethane cuff that surrounds the leg from knee to ankle. The cuff is inflated with air to a pressure of 6 mm Hg and connected to a pressure transducer, amplifier, and chart recorder. Calibration is performed by injecting 100 cc of air into the air chamber. Measurements are made at room temperature.

Technique. The air plethysmograph is fitted with the patient in the supine position, and the leg is elevated on a heel support with the knee slightly bent and externally rotated. The leg is kept in this position for 5 minutes to allow a stable leg/air chamber/room temperature gradient to occur and to ensure resting arterial inflow.

The patient is then asked to stand on a stool with his weight on the opposite leg. The leg with the APG cuff is fully suspended. An increase in leg volume is observed as a result of venous filling. When the chart recorder reaches a plateau, the leg veins are full and this represents the functional venous volume (VV). The venous filling index (VFI) is a measure of the average filling rate of the leg veins while standing and is expressed in milliliters per second. Normal limbs should fill slowly from the arterial inflow.

The patient is then asked to stand on both legs, perform one tiptoe movement, and return to the initial position. The recorded decrease in VV represents the ejection volume (EV) as a result of calf muscle contraction. The ejection fraction (EF) = EV/VV .

After a new plateau is reached the patient is asked to perform 10 tiptoe movements at a rate of one per second and then return to rest. A volume decrease to a new steady state is observed, the residual volume. The residual volume is the amount of blood remaining in the calf at the end of ten tiptoe exercises. The residual volume fraction (RVF) is the percentage of functional VV remaining in the calf. The RVF correlates directly with the ambulatory venous pressure (AVP), that is an RVF of 25% equals an AVP of approximately 25 mm Hg.

The APG was also used for determination of maximal venous outflow (MVO), a test for proximal venous obstruction. With the pneumatic cuff in place, a thigh tourniquet is inflated to 80 mm Hg. An increase in VV to a new plateau (V1) is observed. Once V1 appears the thigh tourniquet is rapidly deflated. The MVO (percent) = $V1/VV$, where V1 is the VV that leaves the leg in 1 second.

Duplex imaging

Equipment. All patients underwent preoperative high-resolution duplex imaging of the deep and

superficial venous system with use of an Ultramark 9 high-definition imaging system with a 5 MHz probe (Advanced Technology Laboratories, Bothell, Wash.).

Technique. Duplex imaging of the deep and superficial venous system was performed with the patient in the supine position. Venous reflux was demonstrated with the patient in 30 degrees of the reverse Trendelenburg position. Venous augmentation was performed by manual compression of the calf or thigh or with the Valsalva technique as described by Hanrahan et al.⁴

Statistics. Statistical analysis of data was performed with use of the Student paired *t* test. The data are represented as the mean $\pm$ 1 SE.

RESULTS

We studied 25 extremities in 23 patients (14 men, 9 women) both before operation and after standard venous surgical operations of the lower extremity by use of APG. Fifteen extremities were classified as having grade I chronic venous insufficiency (primary varicose veins) by physical examination. The remaining 10 extremities had grade III chronic venous insufficiency (severe lipodermatosclerosis with ulceration) when classified according to the standards set by the Society for Vascular Surgery and the North American Chapter of the International Society for Cardiovascular Surgery. Based on preoperative duplex imaging, 14 extremities were found to have incompetence of the superficial venous system only. Eleven extremities were shown to have both superficial and deep venous system incompetence. All patients underwent preoperative APG studies. Follow-up APG studies of the operated extremity were performed in all patients 7 to 10 days after the operation.

Twenty-one extremities underwent ligation and stripping of the GSV (Fig. 1). The GSV was stripped from the groin to the ankle in a standard fashion. The VFI, which averaged 6.6 ± 0.7 ml/sec before operation, improved to a normal level of 1.8 ± 0.3 ml/sec in all patients ($p = 0.0001$). VV decreased from 177.1 ± 14.5 ml to 139.2 ± 8.0 ml after surgery ($p = 0.0008$). EF improved from $45.8\% \pm 2.0\%$ to $50.8\% \pm 2.5\%$ ($p = 0.07$). RVF fell from $45.0\% \pm 3.4\%$ to $42.0\% \pm 3.7\%$ ($p = 0.4$) after surgery.

Four extremities with incompetent popliteal veins and grade III chronic venous insufficiency underwent popliteal vein valve transplantation (Fig. 2). An autogenous axillary vein valve was obtained at the time of surgery and transplanted into the proximal

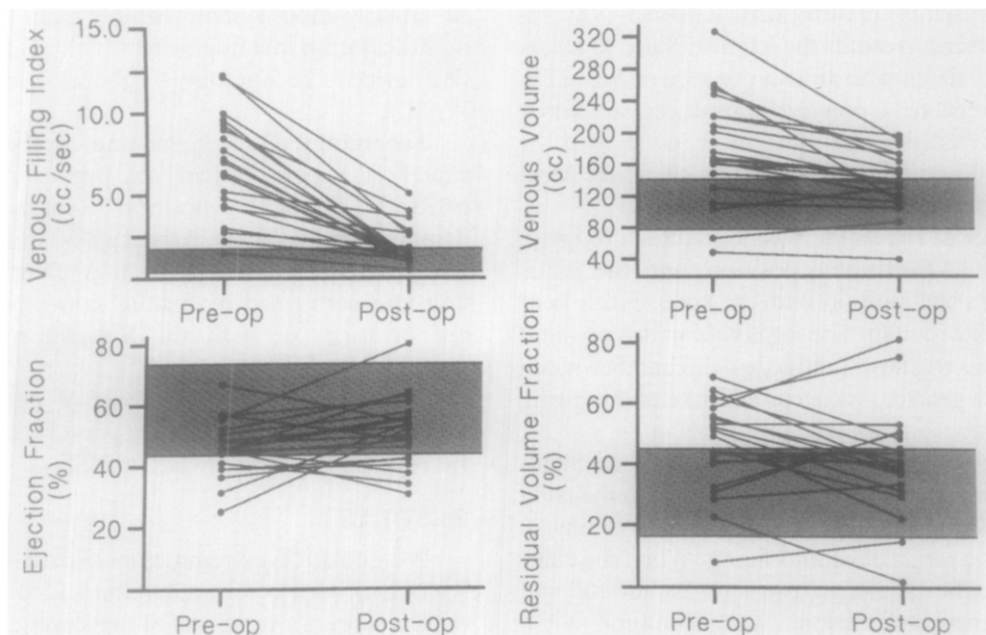


Fig. 1. Graph of change in VV, VFI, EF, and RVF in patients undergoing GSV stripping.

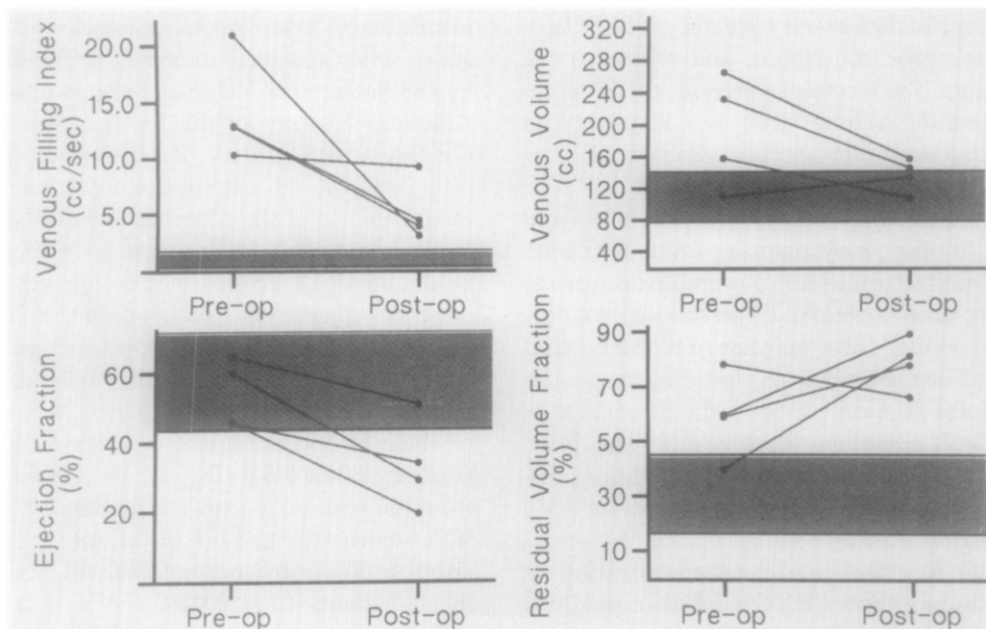


Fig. 2. Graph of change in VV, VFI, EF, and RVF in patients undergoing popliteal vein valve transplantation.

popliteal vein in an anatomic position. No external support devices were used. Postoperative duplex imaging in these four patients revealed that only three of these four valves were patent. One patient had thrombosis in his transplanted valve. In the three

remaining patients VFI decreased dramatically from 14.7 ± 3.2 ml/sec to 5.3 ± 1.9 ml/sec ($p = 0.2$), and VV normalized from 166.2 ± 35.8 ml to 128.2 ± 11.4 ml ($p = 0.4$). EF and RVF worsened in this patient population after popliteal vein valve

transplantation. In subsequent follow-up at 3 months, three of four extremities had improved EF and RVF (Tables I and II).

DISCUSSION

These results show the effectiveness of APG as a technique to monitor the results of venous surgery. In patients undergoing ligation and stripping of the GSV, VFI and VV improved significantly after surgery although they did not always normalize. EF showed a small improvement, whereas RVF showed no consistent pattern of change after surgery. In patients who underwent popliteal vein valve transplantation for deep system incompetence, we found marked reductions in VFI and VV, whereas EF and RVF worsened, at least in the very early postoperative period.

The use of APG as a technique to monitor the results of venous surgery was initially investigated by Christopoulos et al.⁵ In this study APG was used to measure venous hemodynamic changes in patients with primary varicose veins who had flush ligation of the saphenous-femoral junction, saphenous-popliteal junction, or thigh perforating veins. We chose to extend this study further by investigating not only patients with primary varicose veins but also patients with grade III chronic venous insufficiency.

In patients undergoing ligation and stripping of the GSV, the VFI, which was abnormal in 20 of 21 patients before operation, improved significantly after surgery, although it did not normalize in five patients. This may have been a consequence of associated deep venous incompetence not addressed by GSV stripping. Likewise, VV in most patients were abnormal and approached normal after surgery. In this group there was a small improvement in EF of borderline significance. EF normalized in five patients and improved in another 11. The presence of coexisting venous disease may explain why the *p* value for EF did not achieve statistical significance. Similarly, RVF, the estimate of AVP, showed no consistent pattern of change after surgery. Stripping of the GSV, therefore, does not seem to have much effect on AVP.

The results obtained in this study are comparable with results obtained by Christopoulos et al.⁵ for GSV strippings. Their study showed a reduction in mean VV of 16% and mean VFI of 70%. These data closely approximate our study in which the mean VV decreased 21% and mean VFI decreased 72%. On the subject of EF and RVF, however, we were unable to reproduce the improvement in mean EF of 30% and reduction in mean RVF of 46% that these

Table I. EF (expressed as percent) of patients before popliteal vein valve transplantation and at 1-week and 3-month follow-up

Patient	Before operation	1 wk after operation	3 mo after operation
1	45.5	33.8	42.0
2	45.7	19.2	47.5
3	64.7	50.8	61.5
4*	60.0	28.8	28.3

*Patient had thrombosis of transplanted valve after the operation.

Table II. RVF (expressed as percent) of patients before popliteal vein valve transplantation and at 1-week and 3-month follow-up

Patient	Before operation	1 wk after operation	3 mo after operation
1	60	89	41
2	59	77	57
3	78	66	74
4*	39	81	88

*Patient had thrombosis of transplanted valve after the operation.

authors showed. In our study we found the mean EF improved only 9% and mean RVF was reduced by only 7%. The reason for these differences are not clear at this time.

Seven of the 21 patients in this group were found to have deep venous incompetence by duplex scanning. One of these seven patients had a normal VFI before operation. After GSV stripping, three of seven patients had a normal VFI. In these four cases it seems unlikely that popliteal vein incompetence existed. We believe this represents an error in our method of determining venous valvular incompetence. Popliteal valve reflux may be overdiagnosed if reflux velocities greater than 30 cm/sec are not reached with use of the Valsalva or thigh compression techniques, which we use in our vascular laboratory at this time.⁶ We are in agreement with van Bemmel et al.⁶ that a standardized method of producing reflux is needed to reliably establish the presence or absence of reflux in the superficial or deep venous systems.

Popliteal vein valve transplantation was performed in four extremities of three patients who had deep system incompetence (grade IV reflux), who had undergone previous ligation and stripping of their superficial venous system or perforator ligation, and who had recurrent venous ulceration. Preoperative descending venography was performed on two of the four extremities. One patient (two extremities)

refused preoperative venography. Postoperative duplex imaging of the popliteal fossa found three of four transplanted valves to be patent. In the early postoperative period, one patient had a deep venous thrombosis (DVT) involving the valve. In the three extremities without DVT we showed a marked decrease in venous reflux and VV after the operation. The EF and RVF however, had worsened significantly when measured at 1 week after operation. In a study by O'Donnell et al.⁷ it is suggested that EF and RVF should improve after popliteal vein valve transplantation. The deterioration in EF and RVF measurements we found may be explained in part by patient inability to perform tiptoe maneuvers because of pain in the early postoperative period. In the three extremities without DVT, both EF and RVF reverted to preoperative values when measured 3 to 4 months after operation. Although the small number of patients in both of these studies limits our conclusions, the findings suggest that postoperative APG measurements should not be made too early and further study with more patients is needed.

APG offers a unique and quantitative technique to evaluate the results of reconstructive venous surgery. We were able to measure significant changes in VFI and VV after standard venous surgery for varicose veins and chronic venous insufficiency. We were unable to consistently show improvements in EF or RVF in most patients despite, in some cases, dramatic decreases in VFI after corrective venous surgery. Postoperative changes in EF or RVF after these types of surgery may not occur or are simply difficult to detect by this test. Finally, volume changes of the leg measured by APG after corrective venous

surgery do not necessarily reflect changes in lower extremity venous pressure, probably because of venous compliance.

This study has shown APG is simple, quantitative, physiologic, and noninvasive. It has proved to be an invaluable research tool for the investigation of lower extremity venous physiology. We feel this technique represents an improvement on older methods for evaluating the effect of surgery on the venous hemodynamics of the leg.

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